Lecture 4 - 12/11

Thursday, 12 November 2026 13:19

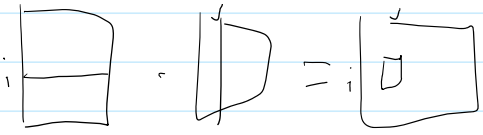
Def: A triangle in a graph $G=(V,E)$ is a triple $\{a,b,c\} \subseteq V$ s.t. $\{a,b\}, \{b,c\}, \{a,c\} \in E$

Δ-detection: given G, does it have a triangle

naive: $O(n^3)$

matrix multiplication
given two $n \times n$ matrices A, B
compute $C = A \cdot B$

$$C[i,j] = \sum_{k=1}^{n} A[i,k] \cdot B[k,j]$$



$$A[i,j] = \begin{cases} 1 & \text{if } \{i,j\} \in E \\ 0 & \text{else} \end{cases}$$

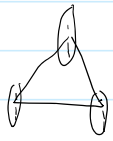
$A^2[i,j]$ = number of 2-paths

find i,j s.t. ① $A^2[i,j] > 0$, ② $(i,j) \in E \equiv A[i,j] = 1$

def: w is the smallest const s.t. MM can be solved in $O(n^w)$ time

Thm: $w < 2.25$

Δ-det (unweighted) $O(n^w)$ then can be tri colored



graph problem ← negative-Δ ⟷ APSP
min-Δ
zero-Δ ↔ conv3sum ⟷ 3sum

negative-Δ: given $G=(V,E,w)$
$w: E \to [-U, U]$, does it contain $\{a,b,c\} \subseteq V$ s.t. $w(a,b) + w(b,c) + w(a,c) < 0$

use weighted MM — $A[i,j] = \begin{cases} w(i,j) & \text{if } \{i,j\} \in \\ \text{int} & \text{else} \end{cases}$

use weighted MM - $A[i,j] = \begin{cases} \text{int} & \text{else} \end{cases}$

$C[i,j] = \min_{k=1}^{n} A[i,k] + B[k,j]$ $\quad C = A \oplus B$

we want $A \oplus A[i,j] + A[i,j] < 0$

MM $(\min,+)$: given two $n \times n$ matrices A, B, compute $C = A \oplus B$ s.t.

$$C[i,j] = \min_{k=1}^{n} A[i,k] + B[k,j]$$

Alg (Ney-G $\to$ $(\min,+)$-MM):

- def A to be the weighted adj matrix

$$A[i,j] = \begin{cases} w(i,j) & \text{if } (i,j) \in E \\ +\infty & \text{else} \end{cases}$$

- compute $A^2 = A \oplus A$
- return $\min_{i,j} w(i,j) + A^2[i,j]$

APSP: Given $G = (V,E)$, w

return dist$[u,v]$ = min weight u to v Path in G) for all $u, v \in V$

$\swarrow$ $(\min,+)$

Thm: If $(\min,+)$-MM is in time $T(n)$ then APSP is in time $O(T(n) \cdot \lg n)$

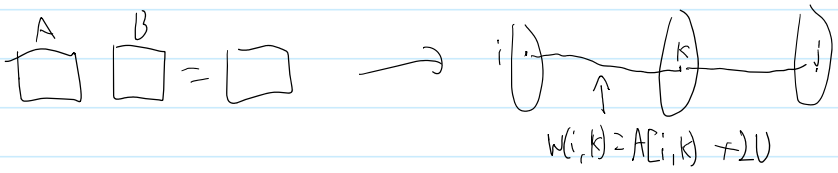
alg: - def A adj weight - $A[i,j] = \begin{cases} w(i,j) & \\ 0 & i = j \\ +\infty & \text{else} \end{cases}$

- compute A^n - via repeated squaring
- return dist$(i,j) = A^n[i,j]$

claim: $A^k[i,j]$ = shortest Path from i to j using $\le k$ edges

time: $\lg n \cdot T(\min,+(n))$

what about $(\min,+)$MM $\to$ APSP



$$dist(i,j) = \min_k A[i,k] + B[k,j]$$

Thm: If Neg-Δ is in time $O(n^{3-\varepsilon})$ then (min,+)-MM (and APSP) in time $O(n^{3-\frac{\varepsilon}{3}})$

APSP $\hookrightarrow$ (min,+) $\longrightarrow$ All pairs Neg-Δ $\longrightarrow$ Neg-Δ

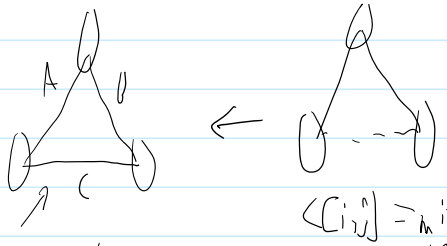
lgn binary search

0/1

All pairs Neg-Δ: given tripartite graph $G=(V,E,w)$

$V = I \cup J \cup K$

return $\forall i \in I, j \in J$ $D[i,j] = \begin{cases} 1 & \exists k \text{ s.t. } w(i,j,k)<0 \\ 0 & \text{else} \end{cases}$



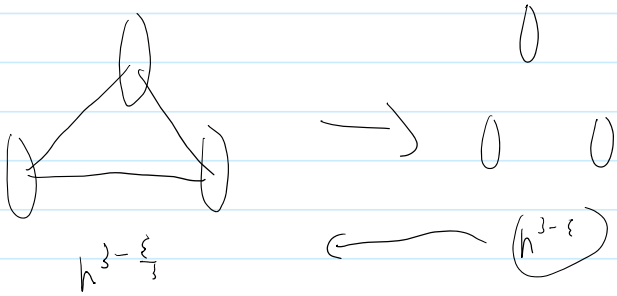
$C[i,j] = \min_k A[i,k]+B[k,j]$

$\forall i,j$ $\exists k$ s.t.

$A[i,k]+B[k,j]$
$+C[i,j] < 0$

done by binary search over all pairs

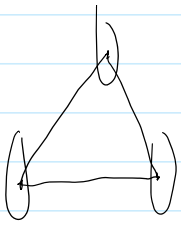
how do we do All pairs neg-Δ $\longrightarrow$ Neg-Δ (finding)



naive: find one Δ at a time $- n^{3-\varepsilon} \cdot n^3$, can be $n^{3-\varepsilon} \cdot n^2$

idea: run det alg on smaller graph on s nodes $s^{3-\varepsilon}$

what smaller s? idea: random subgraphs of size s



$\left(\frac{n}{s}\right)^3$ attempts guarantees that every Δ is formed

$\left(\frac{n}{s}\right)^3 \cdot s^{3-\varepsilon} = \frac{n^3}{s^\varepsilon}$

the problem, when looking for a specific Δ, you might find another

we want to get rid of the collision

Reduction:

split I, J, K arbitrarily into s-sized subsets

$I'_1, \ldots, I'_{n/s}$

$J'_1, \ldots$

$K'_1, \ldots$

- for all $\left(\frac{n}{s}\right)^3$ triples of subset I', J', K' do (in arbitrary order):

⊛ - if Find-Neg-Δ $(I', J', K') = (i, j, k)$

- set $D[i,j] = 1$, remove edge (i,j) from E, set $w(i,j) = +\infty$
- go to ⊛

- o.w. continue to next triple

correctness: if (i,j) is in Δ, we will find it in some triple

time: naive - $\left(\frac{n}{s}\right)^3 \cdot s^{3-\varepsilon} \cdot s^2$

better

$\left[\left(\frac{n}{s}\right)^3 + K\right] \cdot s^{3-\varepsilon}$

K: # pairs (i,j) s.t. $D[i,j]$ in the end $\leq n^2$

so $\left[\left(\frac{n}{s}\right)^3 + n^2\right] \cdot s^{3-\varepsilon} = \frac{n^3}{s^\varepsilon} + n^2 \cdot s^{3-\varepsilon} \overset{s = n^{\frac{1}{3}}}{=} n^{3-\frac{\varepsilon}{3}}$